

Vertex operator algebras for logarithmic conformal field theories

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21 December 2011

Outline

What are log CFT VOAs?

A simple example

Non-semisimple yet well behaved

- ▶ Joint work with Akihiro Tsuchiya from IPMU
- ▶ Results originally conjectured by Matthias Gaberdiel and Ingo Runkel, 2007
- ▶ If I use unfamiliar physics jargon, **please interrupt me.**

Log CFT VOAs

[Gurarie '93]

- ▶ N -point correlation functions contain logarithmic singularities
- ▶ L_0 operator is non-diagonalisable
- ▶ Representation theory is non-semisimple

The $c = -2$ triplet theory

[Kausch '91], [Gaberdiel, Kausch '96], [Adamović, Milas '08]

- ▶ Symplectic fermions
- ▶ Central charge $c = -2$
- ▶ Generated by $T(z)$ and three weight 3 W -fields
- ▶ c_2 -cofinite

So far so good.

The strangeness

Zhu's algebra reveals

$$L^2(L-1)(L+\frac{1}{8})(L-\frac{3}{8})=0.$$

Highest weights are 0 , 1 , $-\frac{1}{8}$ and $\frac{3}{8}$, but something is strange about 0 .

The representation theory

[Nagatomo, Tsuchiya '07]

- ▶ Four irreducibles with weights $0, 1, -\frac{1}{8}, \frac{3}{8}$
- ▶ Four families of extensions of 0 and 1

Can we ignore this strangeness?

Fusion for the $c = -2$ triplet theory

No, we cannot.

$\otimes$	$\mathcal{W}(0)$	$\mathcal{W}(1)$	$\mathcal{W}(-\frac{1}{8})$	$\mathcal{W}(\frac{3}{8})$	$\mathcal{P}(0)$	$\mathcal{P}(1)$
$\mathcal{W}(0)$	$\mathcal{W}(0)$	$\mathcal{W}(1)$	$\mathcal{W}(-\frac{1}{8})$	$\mathcal{W}(\frac{3}{8})$	$\mathcal{P}(0)$	$\mathcal{P}(1)$
$\mathcal{W}(1)$	$\mathcal{W}(1)$	$\mathcal{W}(0)$	$\mathcal{W}(\frac{3}{8})$	$\mathcal{W}(-\frac{1}{8})$	$\mathcal{P}(1)$	$\mathcal{P}(0)$
$\mathcal{W}(-\frac{1}{8})$	$\mathcal{W}(-\frac{1}{8})$	$\mathcal{W}(\frac{3}{8})$	$\mathcal{P}(0)$	$\mathcal{P}(1)$	$2\mathcal{W}(-\frac{1}{8}) \oplus \mathcal{W}(\frac{3}{8})$	$2\mathcal{W}(-\frac{1}{8}) \oplus \mathcal{W}(\frac{3}{8})$
$\mathcal{W}(\frac{3}{8})$	$\mathcal{W}(\frac{3}{8})$	$\mathcal{W}(-\frac{1}{8})$	$\mathcal{P}(1)$	$\mathcal{P}(0)$	$2\mathcal{W}(-\frac{1}{8}) \oplus \mathcal{W}(\frac{3}{8})$	$2\mathcal{W}(-\frac{1}{8}) \oplus \mathcal{W}(\frac{3}{8})$
$\mathcal{P}(0)$	$\mathcal{P}(0)$	$\mathcal{P}(1)$	$2\mathcal{W}(-\frac{1}{8}) \oplus \mathcal{W}(\frac{3}{8})$	$2\mathcal{W}(-\frac{1}{8}) \oplus \mathcal{W}(\frac{3}{8})$	$2\mathcal{P}(0) \oplus 2\mathcal{P}(1)$	$2\mathcal{P}(0) \oplus 2\mathcal{P}(1)$
$\mathcal{P}(1)$	$\mathcal{P}(1)$	$\mathcal{P}(0)$	$2\mathcal{W}(-\frac{1}{8}) \oplus \mathcal{W}(\frac{3}{8})$	$2\mathcal{W}(-\frac{1}{8}) \oplus \mathcal{W}(\frac{3}{8})$	$2\mathcal{P}(0) \oplus 2\mathcal{P}(1)$	$2\mathcal{P}(0) \oplus 2\mathcal{P}(1)$

Current Lie algebra

[Matsuo, Nagatomo, Tsuchiya '05]

Let V be a c_2 -cofinite VOA with vertex operator map

$$Y : V \rightarrow \text{End}(V)[[z, z^{-1}]]$$

$$A \mapsto \sum_{n \in \mathbb{Z}} A_n z^{-n-h}.$$

We denote the Lie algebra of modes by $\mathfrak{g}(V)$

$$A_n = A \otimes z^{n+h-1} dz^{1-h}.$$

Defining the fusion product $\dot{\otimes}$

[Gaberdiel '93], [Nahm '94], [Hashimoto, Tsuchiya]

The action of $A \otimes f(z)dz^{1-h}$, $f(z) \in \mathbb{C}[z, z^{-1}, (z-1)^{-1}]$ on $M \otimes N$ is given by

$$\rho_{1,0}(A \otimes f(z)dz^{1-h}) = (A \otimes f_1(\xi_1)d\xi_1^{1-h}) \otimes 1 + 1 \otimes (A \otimes f_0(\xi_0)d\xi_0^{1-h}).$$

Identify the Lie algebra element this action corresponds to by

$$\rho_{\infty}(A \otimes f(z)dz^{1-h}) = A \otimes f_{\infty}(\xi_{\infty})d\xi_{\infty}^{1-h}.$$

Defining the fusion product $\dot{\otimes}$

[Gaberdiel '93], [Nahm '94], [Hashimoto, Tsuchiya]

In order to analyse $M \dot{\otimes} N$ more easily define quotients

$$\frac{M \otimes N}{\mathfrak{g}_{k+1}(V)(M \otimes N)} ,$$

where $A \otimes f(z)dz^{1-h} \in \mathfrak{g}_k(V)$ if the leading order of $f(z)$ is less than $h - 1 - k$. Then

$$\overline{M \dot{\otimes} N} = \varprojlim_k \frac{M \otimes N}{\mathfrak{g}_{k+1}(V)(M \otimes N)} .$$

Resulting product structure

[Hashimoto, Tsuchiya]

The fusion product is given by

$$M \dot{\otimes} N = \bigoplus_h \overline{M \dot{\otimes} N}[h] .$$

Theorem (Hashimoto, Tsuchiya)

This definition of the fusion product works for non-semisimple representation categories and endows $V\text{-mod}$ with the structure of a braided monoidal category.

Main result

[Tsuchiya, SW]

The $c = -2$ triplet theory is the simplest example of the $\mathcal{W}_p$ triplet series.

Theorem

- ▶ $\mathcal{W}_p\text{-mod}$ has the structure of a rigid braided monoidal category.
- ▶ Fusion closes on the set of all simple and all projective modules, but not just on the simple modules.

Summary and outlook

Conclusions:

- ▶ Log CFT VOAs can arise from innocent-looking c_2 -cofinite VOAs
- ▶ Rich representation theory

Open problems:

- ▶ More complicated log CFT VOAs such as the $\mathcal{W}_{p,q}$ -series
- ▶ Representation category is braided monoidal but not necessarily rigid
- ▶ Need more powerful tools for classifying all indecomposables